

**LUNCH INTRODUCTIONS  
PAUL “CHIP” JAENICHEN  
ACTING MARITIME ADMINISTRATOR**

**NATIONAL MARITIME STRATEGY SYMPOSIUM #1  
*GROWING THE US-FLAG FLEET ENGAGED IN INTERNATIONAL TRADE***

**JANUARY 15, 2014**

**U.S. DEPARTMENT OF TRANSPORTATION (DOT)  
WEST ATRIUM – GROUND FLOOR  
WASHINGTON, DC**

**BEFORE WE WELCOME OUR NEXT SPEAKER, I  
HAVE A BIT OF NEWS.**

**UNFORTUNATELY, SOME LAST MINUTE CHANGES  
TO THE HOUSE JUDICIARY COMMITTEE SCHEDULE  
ARE PREVENTING CONGRESSMAN RANDY FORBES  
FROM JOINING US TODAY.**

**HE SENDS HIS APOLOGIES AND BEST WISHES FOR  
OUR EFFORT.**

**ALSO, ALTHOUGH WE'D INITIALLY SCHEDULED MR  
FOTIS KARAMITSOS (*FOE-TISS CAR-AH-MEAT-  
ZOHS*), TO SPEAK TODAY, AN URGENT FAMILY  
ISSUE REQUIRES THAT HE BE ELSEWHERE RIGHT  
NOW.**

**HOWEVER, WE HAVE BEEN VERY FORTUNATE TO  
HAVE TWO KNOWLEDGEABLE SPEAKERS COMMIT  
ON SHORT NOTICE . . .**

**FIRST, WE HAVE WITH US A MAN WITH A WELL-  
ROUNDED AND DEEP KNOWLEDGE OF U.S.  
TRANSPORTATION POLICY.**

**HE SPENT NINE YEARS ON THE HILL – AS A STAFF  
MEMBER FOR THE SENATE COMMITTEE ON  
COMMERCE, SCIENCE AND TRANSPORTATION . . .  
AND THE HOUSE COMMITTEE ON TRANSPORTATION  
INFRASTRUCTURE.**

**AND NOW HE’S A MAJOR ASSET TO US HERE AT  
THE D.O.T.**

**HE CURRENTLY SERVES IN THE OFFICE OF THE  
ASSISTANT SECRETARY FOR POLICY HERE AT THE  
DEPARTMENT, WHERE HE IS RESPONSIBLE FOR  
COORDINATING THE DEPARTMENT’S NATIONAL  
FREIGHT ACTIVITIES . . . INCLUDING MANAGING THE  
NATIONAL FREIGHT ADVISORY COMMITTEE.**

**HE ALSO OVERSEES THE DEVELOPMENT OF THE  
DEPARTMENT'S UPCOMING NATIONAL FREIGHT  
STRATEGIC PLAN AND THE IMPLEMENTATION OF  
THE NATIONAL FREIGHT POLICY.**

**PLEASE HELP ME WELCOME JOHN DRAKE!**

**. . .**

**. . .**

**. . .**

**THANK YOU, JOHN . . .**

**IT WAS OUR INTENTION TO WIDEN HORIZONS WITH  
THIS SYMPOSIUM . . . AND GET MARITIME PLAYERS  
TO TAKE A LOOK AT THE BIGGER PICTURE IN THIS  
INDUSTRY.**

**THAT'S WHY IT IS SUCH AN HONOR TO INTRODUCE  
OUR NEXT SPEAKER . . . A PERSON WHO CAN SHINE**

**SOME LIGHT ON MARITIME PRACTICES AND DEVELOPMENTS ACROSS THE ATLANTIC.**

**HE HAS BEEN WITH THE EUROPEAN COMMISSION SINCE 2003, PLAYING A LEADING ROLE IN NUMEROUS RESEARCH, LOGISTICS AND ANALYSIS EFFORTS FOCUSING ON EUROPEAN INLAND WATERWAY TRANSPORT AND MARITIME SHIPPING...**

**AMONG THESE, HE WAS A CENTRAL CONTRIBUTOR TO THE EUROPEAN COMMISSION'S 2011 WHITE PAPER ON TRANSPORT, A POLICY STRATEGY TO SERVE THE E.U. THROUGH 2021 . . .**

**PLEASE JOIN ME IN GIVING A WARM WELCOME TO TRANSPORT COUNSELOR FOR THE EUROPEAN UNION DELEGATION TO THE UNITED STATES, DR FELIX LEINENMAN (*LINE-E-MAHN*).**

# TALC AND PYROPHYLLITE

By Robert L. Virta

**Domestic survey data and tables were prepared by Christopher H. Lindsay, statistical assistant, and the world production table was prepared by Glenn J. Wallace, international data coordinator.**

The mineral talc is a hydrous magnesium silicate. A massive talcose rock is called steatite, and an impure massive variety is known as soapstone. Talc is used commercially because of its fragrance retention, luster, purity, softness, and whiteness. Other commercially important properties of talc are its chemical inertness, high dielectric strength, high thermal conductivity, low electrical conductivity, and oil and grease adsorption. Major markets for talc are ceramics, paint, paper, and plastics.

Pyrophyllite is a hydrous aluminum silicate with a structure similar to talc. Such properties as chemical inertness, high dielectric strength, high melting point, and low electrical conductivity make it useful for ceramic and refractory applications.

## Legislation and Government Programs

In 1999, the U.S. Department of Defense authorized the disposal of 907 metric tons (t) of block and lump talc from the National Defense Stockpile, which is the entire uncommitted inventory in that category; 2 t was sold.

## Production

**Talc.**—In 1999, 9 companies operating 13 mines in 6 States produced soapstone, steatite, and talc. All but one operation were open pit mines. The producers were, in decreasing order of production, Luzenac America Inc., Barrett's Minerals Inc., Dal Minerals Co., Gouverneur Talc Co., Milwhite Inc., Suzorite Mineral Products Inc. (Zemex Corp.), Unimin Texas Co. L.P. (formerly United Clays Inc. of Texas), Steatite of Southern Oregon, and NuTECH Minerals Inc. Barrett's Minerals, Dal Minerals, Gouverneur Talc, and Luzenac America were the largest domestic producers, accounting for more than 80% of the tonnage.

In 1999, U.S. mine production was 925,000 t valued at \$26.1 million, compared with 971,000 t valued at \$27.3 million in 1998 (tables 1 and 2). Production increased slightly in New York; decreased in California, Montana, Texas, and Vermont; and remained unchanged in Oregon. Montana led all States in the tonnage and value of talc produced, followed by Texas, Vermont, New York, Oregon, and California. Mines that operated in Montana, New York, Texas, and Vermont accounted for nearly all the domestic talc production.

Domestic production data were obtained through a voluntary U.S. Geological Survey (USGS) survey of U.S. mining companies. Survey forms were sent to 10 companies with responses accounting for approximately 80% of the data

presented in table 1; the remainder was estimated from reported prior-year data adjusted according to industry trends.

**Pyrophyllite.**—Standard Industrial Minerals Inc. operated one mine in California, and Piedmont Minerals Co. and Standard Mineral Co. Inc. operated three mines in North Carolina. Production of pyrophyllite decreased from that of 1998.

Domestic production data were acquired through a voluntary USGS survey of the three U.S. companies that mine pyrophyllite. All responded to the survey.

## Consumption

**Talc.**—Approximately 881,000 t of talc valued at \$102 million was sold or used in 1999, an increase from 870,000 t valued at \$109 million in 1998. Sales or use of talc increased in Montana and Texas; decreased in California, New York, and Vermont; and remained essentially unchanged in Oregon. Of the 881,000 t of talc sold or used, producers reported that 771,000 t was sold for domestic use and approximately 110,000 t was exported.

Domestic markets included, in decreasing order of consumption, ceramics (pottery, sanitaryware, tiles, etc.), paper, paint, plastics, roofing, cosmetics, rubber, insecticides, and refractories (table 3). Domestic sales of talc did not change significantly between 1998 and 1999.

About 73,000 t of talc was reported under the "Other" category by respondents in 1999. Of this amount, 55,600 t was used in animal feed additives, automobile body fillers, caulks, joint compounds, paint and putties, sculpture media, tile flooring, and vinyl sheet flooring. The remainder of the "Other" category (17,400 t) was used in applications that were not identified by respondents.

Most of the imported talc was not included in the domestic end-use data listed in table 3. More than 80% of the imported talc was purchased by mineral brokers who do not participate in the USGS canvass. An estimate of the end-use breakdown based on countries of origin, ports of entry, importing companies, and regional end-use patterns is ceramics and refractories, 15,000 t; cosmetics, 8,000 t; paint, 31,000 t; paper, 12,000 t; plastics, 80,000 t; rubber, 9,000 t; and unknown, 23,000 t.

**Pyrophyllite.**—Domestic consumption of pyrophyllite decreased from that of 1998. Pyrophyllite was used, in decreasing order of consumption, in ceramics, refractories, paint, plastics, insecticides, and rubber. Sales increased for paint and plastics applications and decreased for the others. Ceramic and refractory uses accounted for 61% of domestic

pyrophyllite sales.

Domestic consumption data for talc and pyrophyllite were developed by the USGS from a voluntary survey of U.S. mills. Survey forms were sent to 11 companies operating 16 mills in 8 States for talc and 3 companies operating 3 mills in 2 States for pyrophyllite. Approximately 87% of the talc data presented in table 3 was reported by the companies; the remainder was estimated from reported prior-year data adjusted according to industry trends. All the pyrophyllite producers responded to the consumption survey.

## Prices

Talc prices varied depending on the quality and the degree and method of processing. The unit value of crude talc was estimated to be \$28 per metric ton. More than 80% of the crude ore value included in table 1 was estimated because most producers do not sell crude talc and could not provide a crude ore value. The average reported unit value of processed talc was \$116 per ton. The average unit values of crude and processed pyrophyllite increased slightly in 1999.

The average unit value for exports of unmilled talc was \$114 per ton, and that of milled talc was \$195 per ton. The average unit value for imports was \$83 per ton for unground talc, \$161 per ton for ground talc, and \$932 per ton for cut or sawed talc.

Bidding for export licenses on 250,000 t of Chinese talc took place in March 1999. Licenses, in general, increased about \$8 per ton to \$20 per ton. Industry experts believe that much of the increase in licensing costs could be absorbed through cost reductions by Chinese talc producers (Mineral PriceWatch, 1999).

Approximate prices for talc ranged from \$83 to \$371 per ton (table 4; Industrial Minerals, 1999d). Quoted prices should be used only as a guideline because actual prices depend on the terms of the contract between seller and buyer.

## Foreign Trade

Talc exports increased slightly in tonnage, to 147,000 t, and 5% in value, to \$27.2 million. Canada was the leading importer of U.S. talc, followed by Mexico, Belgium, Japan, Venezuela (7,220 t), and Taiwan (4,560 t) (table 5). Since 1990, data reported by domestic producers concerning exports to Mexico gradually have diverged from the Bureau of the Census data. On the basis of industry data, exports to Mexico are estimated to be more than three times those reported by the Bureau of the Census.

Talc imports reported by the Bureau of the Census increased by 26% in tonnage, to 208,000 t, and increased by 52% in value, to \$35.3 million. Such a large tonnage increase probably could not be absorbed by current talc markets and it is likely that much was processed and stockpiled for future sales. Cut and sawed talc, which has a unit value of \$932 per ton, contributed disproportionately to the large increase in the import value compared with import values for unmilled and milled talc. Canada, China, France, and Japan supplied 89% of

all talc imports (table 6). Some of the talc imported from Japan was likely to have been transshipments from other Southeast Asian countries or Australia.

Approximately 50% (103,000 t) of talc imported into the United States came through the port district of New Orleans, LA, almost all unmilled. Of this amount, 6,700 t was imported from Australia; 86,700 t from China; 68 t from France; 8,990 t from Hong Kong; and 244 t from Italy. The second leading port district was Detroit, MI, with 25,400 t (principally from Canada); followed by Boston, MA, with 21,000 t (principally from France); and Buffalo, NY, with 12,400 t (principally from Canada). The remaining imports were scattered among the many other port districts.

## World Review

China remained the world's leading producer of talc, followed by the United States, Brazil, India, Finland, and France. The Republic of Korea was the largest producer of pyrophyllite, followed by Japan and Brazil. China, Japan, the Republic of Korea, and the United States produced 69% of the world's talc and pyrophyllite (table 7).

**Australia.**—WMC Resources Ltd. announced plans to build an \$11 million mill with a capacity of 35,000 to 40,000 metric tons per year (t/yr) at either its Three Springs or Kwinana site. The mill will permit WMC to use talc from its Three Springs mine for higher value applications. The paper and paint markets in Australia and Southeast Asia are being served by the company's Amsterdam plant (Industrial Minerals, 1999e).

**Canada.**—Highwood Resources Ltd. acquired a talc mine and mill from Canada Talc Ltd. The mine and mill are located in Madoc, Ontario. Highwood Resources is beginning underground development at the mine to improve access to higher grade ore. The company also is upgrading the mill facilities (Industrial Minerals, 1999b).

Trinity Resources and Energy Ltd. began preparing its pyrophyllite mine and mill, which have been idle since 1995, for operation. In 1998, the company acquired the mine and mill, located in Newfoundland, from Armstrong World Industries Canada Ltd. The purchase included the mine, mill, 1 million tons of reserves, and ceramic- and mixed-grade stocks. The company expects to be fully operational in 2000 and intends to diversify from ceramics into pigment and filler markets, such as paint, plastic, and roofing (North American Minerals News, 1999).

**Italy.**—Luzenac Val Chisone SpA opened an underground mine in Rodoretto. The deposit contains high-purity, white, macrocrystalline ore. Luzenac anticipates that production will be 60,000 t/yr. The talc will be sold for polymer, cosmetic, pharmaceutical, and coating applications (Industrial Minerals, 1999c).

**Mexico.**—Luzenac Europe acquired 51% ownership of Sierra Talc de Mexico, SA de CV. Sierra Talc is a toll grinding operation with an average output of 40,000 t/yr. It has mines in Guerrero and Puebla and mills in Monterrey and Mexico City. The company mines about 10% of its feed

material with the remainder obtained from other producers (Industrial Minerals, 1999a).

## Outlook

Markets for talc should be stable and even slightly increasing for the next few years. Major markets for talc, such as ceramics, paint, paper, and plastics, are projected to grow slowly through 2003 as a result of the strong U.S. economy. The plastics market should continue to offer the greatest growth potential for talc, and imports will play a larger role in the U.S. talc market. No major changes are anticipated in pyrophyllite markets, and consumption probably will not change significantly for the next few years.

## References Cited

- Industrial Minerals, 1999a, ...And acquires majority stake in Sierra Talc: Industrial Minerals, no. 378, March, p. 9.  
———1999b, Canada Talc bought by Highwood Resources: Industrial Minerals, no. 378, March, p. 9.  
———1999c, Luzenac opens talc mine in Italy: Industrial Minerals, no. 378, March, p. 21.  
———1999d, Prices: Industrial Minerals, no. 387, December, p. 71.  
———1999e, WMC to build new talc mill: Industrial Minerals, no. 381, June, p. 106.  
Mineral PriceWatch, 1999, Talc—Second round results released: Mineral

PriceWatch, no. 52, April, p. 2.  
North American Minerals News, 1999, Trinity Resources and Energy: North American Minerals News, no. 44, January, p. 7-8.

## GENERAL SOURCES OF INFORMATION

### U.S. Geological Survey Publications

- Talc and pyrophyllite. Ch. in Mineral Commodity Summaries, annual.<sup>1</sup>  
Talc. Ch. in United States Mineral Resources, Professional Paper 820, 1973.

### Other

- Ceramic Industry.  
Engineering and Mining Journal.  
Mining Engineering.  
Paint and Coatings Industry.  
Talc and pyrophyllite. Ch. in Mineral Facts and Problems, U.S. Bureau of Mines Bulletin 675, 1985.  
Talc. The Talc Industry—An Overview, U.S. Bureau of Mines Information Circular 9220, 1989.

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<sup>1</sup>Prior to January 1996, published by the U.S. Bureau of Mines.

TABLE 1  
SALIENT TALC AND PYROPHYLLITE STATISTICS 1/

(Thousand metric tons, unless otherwise specified)

|                                         | 1995     | 1996      | 1997      | 1998        | 1999      |
|-----------------------------------------|----------|-----------|-----------|-------------|-----------|
| United States:                          |          |           |           |             |           |
| Mine production, crude:                 |          |           |           |             |           |
| Talc                                    | 1,060    | 994       | 1,050     | 971 r/      | 925       |
| Pyrophyllite                            | W        | W         | W         | W           | W         |
| Total                                   | 1,060    | 994       | 1,050     | 971 r/      | 925       |
| Value:                                  |          |           |           |             |           |
| Talc thousands                          | \$31,700 | \$31,100  | \$33,000  | \$27,300 r/ | \$26,100  |
| Pyrophyllite do.                        | W        | W         | W         | W           | W         |
| Total do.                               | \$31,700 | \$31,100  | \$33,000  | \$27,300 r/ | \$26,100  |
| Sold by producers, crude and processed: |          |           |           |             |           |
| Talc                                    | 901      | 909       | 942       | 870 r/      | 881       |
| Pyrophyllite                            | W        | W         | W         | W           | W         |
| Total                                   | 901      | 909       | 942       | 870 r/      | 881       |
| Value:                                  |          |           |           |             |           |
| Talc thousands                          | \$99,900 | \$100,000 | \$111,000 | \$109,000   | \$102,000 |
| Pyrophyllite do.                        | W        | W         | W         | W           | W         |
| Total do.                               | \$99,900 | \$100,000 | \$111,000 | \$109,000   | \$102,000 |
| Exports (talc) 2/                       | 183      | 192       | 179       | 146         | 147       |
| Value thousands                         | \$37,100 | \$37,900  | \$34,200  | \$26,000    | \$27,200  |
| Imports for consumption                 | 146      | 187       | 123       | 165         | 208       |
| Value thousands                         | \$14,800 | \$20,500  | \$21,100  | \$23,300    | \$35,300  |
| Apparent consumption 3/                 | 1,020    | 989       | 992       | 1,080       | 986       |
| World, production                       | 8,490 r/ | 9,880 r/  | 10,400 r/ | 9,430 r/    | 9,470 e/  |

e/ Estimated. r/ Revised. W Withheld to avoid disclosing company proprietary data.

1/ Data are rounded to no more than three significant digits; may not add to totals shown.

2/ Excludes powders--talcum (in package), face, and compact.

3/ Production plus imports minus exports plus adjustments in Government and industry stock. Does not include pyrophyllite.

TABLE 2  
CRUDE TALC PRODUCED IN THE UNITED STATES, BY STATE 1/ 2/

(Thousand metric tons and thousand dollars)

| State    | 1998 r/  |        | 1999     |        |
|----------|----------|--------|----------|--------|
|          | Quantity | Value  | Quantity | Value  |
| Texas    | 245      | 5,230  | 220      | 5,000  |
| Other 3/ | 725      | 22,000 | 705      | 21,100 |
| Total    | 971      | 27,300 | 925      | 26,100 |

r/ Revised.

1/ Data are rounded to no more than three significant digits; may not add to totals shown.

2/ Excludes pyrophyllite.

3/ Includes California, Montana, New York, Oregon, and Vermont.

TABLE 3  
END USES FOR GROUND TALC 1/

(Thousand metric tons)

|              | 1998   | 1999 |
|--------------|--------|------|
| Ceramics     | 212 r/ | 209  |
| Cosmetics    | 32 r/  | 27   |
| Insecticides | 10 r/  | 8    |
| Paint        | 135 r/ | 142  |
| Paper        | 160 r/ | 167  |
| Plastics     | 53 r/  | 64   |
| Refractories | 5      | 2    |
| Roofing      | 55 r/  | 60   |
| Rubber       | 23 r/  | 19   |
| Other 2/     | 78 r/  | 73   |
| Total        | 764 r/ | 771  |

r/ Revised.

1/ Excludes pyrophyllite.

2/ Includes art sculpture, asphalt filler, auto body filler, construction caulks, flooring, joint compounds, and other uses not specified.

TABLE 4  
PRICES OF TALC

(U.S. dollars per metric ton)

|                             |         |
|-----------------------------|---------|
| New York:                   |         |
| Paint:                      |         |
| 200 mesh                    | 100     |
| 400 mesh                    | 180     |
| Ceramic:                    |         |
| 200 mesh                    | 83      |
| 325 mesh                    | 92      |
| Italian, cosmetic-grade     | 288     |
| Chinese, normal (ex-store): |         |
| UK 200 mesh                 | 321-363 |
| UK 350 mesh                 | 338-371 |

Source: Industrial Minerals, December 1999.

TABLE 5  
U.S. EXPORTS OF TALC 1/ 2/

(Thousand metric tons and thousand dollars)

| Country   | 1998     |        | 1999     |        |
|-----------|----------|--------|----------|--------|
|           | Quantity | Value  | Quantity | Value  |
| Belgium   | 15       | 1,960  | 14       | 1,780  |
| Canada 3/ | 45       | 8,270  | 55       | 9,910  |
| Japan     | 6        | 1,000  | 7        | 1,250  |
| Mexico    | 18       | 2,290  | 17       | 1,780  |
| Other 4/  | 61       | 12,500 | 54       | 12,500 |
| Total     | 146      | 26,000 | 147      | 27,200 |

1/ Data are rounded to no more than three significant digits; may not add to totals shown.

2/ Excludes powders-talcum (in package), face, and compact.

3/ Probably includes shipments in transit through Canadian ports.

4/ Includes 62 countries in 1998 and 47 countries in 1999.

Source: Bureau of the Census.

TABLE 6  
U.S. IMPORTS FOR CONSUMPTION OF TALC, BY COUNTRY 1/

| Country  | Not crushed or powdered |                   | Crushed or powdered    |                   | Cut and sawed          |                   | Total unmanufactured   |                   |
|----------|-------------------------|-------------------|------------------------|-------------------|------------------------|-------------------|------------------------|-------------------|
|          | Quantity (metric tons)  | Value (thousands) | Quantity (metric tons) | Value (thousands) | Quantity (metric tons) | Value (thousands) | Quantity (metric tons) | Value (thousands) |
| 1998:    |                         |                   |                        |                   |                        |                   |                        |                   |
| Brazil   | 6                       | \$6               | 114                    | \$28              | 928                    | \$824             | 1,050                  | \$858             |
| Canada   | 19                      | 8                 | 29,700                 | 6,640             | 84                     | 108               | 29,800                 | 6,750             |
| China    | 68,400                  | 5,010             | 7,850                  | 132               | 732                    | 515               | 77,000                 | 5,660             |
| France   | 6,060                   | 656               | 28,000                 | 313               | 5,670                  | 5,700             | 39,700                 | 6,670             |
| Japan    | --                      | --                | 5,820                  | 538               | --                     | --                | 5,820                  | 538               |
| Other 2/ | 10,200                  | 894               | 1,450                  | 402               | 519                    | 1,520             | 12,200                 | 2,820             |
| Total    | 84,700                  | 6,580             | 72,900                 | 8,050             | 7,930                  | 8,660             | 165,000                | 23,300            |
| 1999:    |                         |                   |                        |                   |                        |                   |                        |                   |
| Brazil   | 2                       | 5                 | 75                     | 19                | 1,230                  | 1,070             | 1,310                  | 1,090             |
| Canada   | 89                      | 37                | 44,100                 | 11,400            | 6,220                  | 4,980             | 50,400                 | 16,500            |
| China    | 91,900                  | 7,380             | 1,070                  | 251               | 579                    | 387               | 93,500                 | 8,020             |
| France   | 175                     | 235               | 21,300                 | 384               | 5,060                  | 5,180             | 26,600                 | 5,800             |
| Japan    | --                      | --                | 15,100                 | 869               | 6                      | 19                | 15,100                 | 888               |
| Other 2/ | 20,000                  | 1,600             | 892                    | 357               | 508                    | 1,040             | 21,400                 | 3,000             |
| Total    | 112,000                 | 9,260             | 82,600                 | 13,300            | 13,600                 | 12,700            | 208,000                | 35,300            |

-- Zero.

1/ Data are rounded to no more than three significant digits; may not add to totals shown.

2/ Includes 26 countries in 1998 and 18 countries in 1999.

Source: Bureau of the Census.

TABLE 7  
TALC AND PYROPHYLLITE: WORLD PRODUCTION, BY COUNTRY AND PRODUCT 1/ 2/

(Metric tons)

| Country 3/                                     | 1995       | 1996         | 1997          | 1998          | 1999 e/    |
|------------------------------------------------|------------|--------------|---------------|---------------|------------|
| Argentina:                                     |            |              |               |               |            |
| Pyrophyllite                                   | 4,189      | 2,180        | 2,242 r/      | 4,000 r/ e/   | 4,000      |
| Steatite e/                                    | 300        | 300          | 300           | 300           | 300        |
| Talc                                           | 12,474     | 11,777       | 13,380 r/     | 13,500 r/     | 13,000     |
| Australia: e/                                  |            |              |               |               |            |
| Pyrophyllite                                   | 5,000      | 5,000        | 5,000         | 5,000         | 5,000      |
| Talc                                           | 210,000    | 210,000      | 210,000       | 210,000       | 210,000    |
| Austria: Steatite                              | 131,614    | 130,000 e/   | 155,730       | 156,000 e/    | 150,000    |
| Brazil: Unspecified 4/                         | 450,684    | 452,180      | 452,000 e/    | 452,000 e/    | 452,000    |
| Canada: Pyrophyllite, soapstone, talc          | 116,000    | 77,000       | 73,000 r/     | 78,000 r/     | 79,000     |
| Chile: Talc                                    | 4,107      | 4,276        | 3,986         | 3,772 r/      | 3,800      |
| China: Unspecified e/                          | 2,400,000  | 4,000,000 r/ | 4,100,000 r/  | 3,800,000 r/  | 3,900,000  |
| Colombia: Pyrophyllite, soapstone, talc        | 19,248     | 14,800 e/    | 14,832        | 15,000 r/ e/  | 15,000     |
| Egypt: Pyrophyllite, soapstone, steatite, talc | 38,608 r/  | 41,227 r/    | 43,627 r/     | 39,720 r/     | 40,000     |
| Finland: Talc                                  | 464,000    | 345,000      | 350,000 e/    | 350,000 e/    | 350,000    |
| France: Talc, crude                            | 322,300    | 349,270      | 350,000 e/    | 325,000 e/    | 350,000    |
| Germany: Talc (marketable)                     | 14,170     | 10,005       | 8,819         | 15,473 r/     | 15,000     |
| Greece: Steatite e/                            | 500        | --           | --            | --            | --         |
| Hungary: Talc e/                               | 1,150      | 1,200        | 1,200         | 1,200         | 1,200      |
| India:                                         |            |              |               |               |            |
| Pyrophyllite                                   | 131,137    | 143,172      | 121,566 r/    | 79,951 r/     | 85,000     |
| Steatite                                       | 469,692    | 472,001      | 417,613 r/    | 447,550 r/    | 450,000    |
| Iran: Talc e/ 5/                               | 20,000 r/  | 20,000 r/    | 20,000 r/     | 20,000 r/     | 20,000     |
| Italy: Steatite and talc e/                    | 136,000    | 168,000      | 142,000       | 140,000       | 140,000    |
| Japan:                                         |            |              |               |               |            |
| Pyrophyllite                                   | 947,713    | 913,973      | 913,822       | 764,079 r/    | 750,000    |
| Talc                                           | 57,269     | 56,153       | 53,000 e/     | 50,000 r/ e/  | 50,000     |
| Korea, North: Unspecified e/                   | 180,000    | 180,000      | 180,000       | 150,000       | 120,000    |
| Korea, Republic of:                            |            |              |               |               |            |
| Pyrophyllite                                   | 789,994    | 780,062      | 994,366       | 843,609 r/    | 850,000    |
| Talc                                           | 29,364     | 19,066       | 25,751        | 24,411 r/     | 25,000     |
| Macedonia: Talc e/                             | 10,000     | 10,000       | 10,000        | 10,000        | 10,000     |
| Mexico: Talc                                   | 11,134     | 10,100 r/    | 13,586        | 18,843 r/     | 20,000     |
| Morocco                                        | 8,429 r/   | 13,053 r/    | 19,850 r/     | 20,000 r/     | 20,000     |
| Nepal: Talc 6/                                 | 1,500 e/   | 5,323        | 6,809         | 6,500 e/      | 6,500      |
| Norway: Talc e/                                | 30,000     | 30,000       | 30,000        | 28,000        | 26,000     |
| Pakistan: Pyrophyllite                         | 35,043     | 34,095       | 45,414 r/     | 48,927 r/     | 50,000     |
| Paraguay: Unspecified e/                       | 200        | 200          | 200           | 200           | 200        |
| Peru: e/                                       |            |              |               |               |            |
| Pyrophyllite                                   | 8,000      | 8,000        | 8,000         | 8,000         | 8,000      |
| Talc                                           | 13,818 7/  | 12,985 7/    | 13,000        | 13,000        | 13,000     |
| Portugal: Talc                                 | 8,400 e/   | 8,277        | 8,236 r/      | 8,400 e/      | 8,400      |
| Romania: Talc                                  | 9,976      | 10,248       | 7,578         | 8,000 e/      | 8,000      |
| Russia: Talc e/                                | 100,000    | 100,000      | 90,000        | 90,000        | 90,000     |
| South Africa:                                  |            |              |               |               |            |
| Pyrophyllite                                   | 5,519      | 2,140        | 2,129         | 2,532 r/      | 2,606 7/   |
| Talc                                           | 9,173      | 16,397       | 24,400 r/     | 21,900 r/     | 18,249 7/  |
| Spain: Steatite                                | 112,341 r/ | 109,756 r/   | 110,000 r/ e/ | 110,000 r/ e/ | 110,000    |
| Sweden: Talc e/                                | 25,000     | 30,000 7/    | 25,000        | 25,000        | 25,000     |
| Taiwan: Talc                                   | 3,500      | 1,500        | 1,331         | 73 r/         | 100        |
| Thailand:                                      |            |              |               |               |            |
| Pyrophyllite                                   | 76,189     | 64,330       | 304,524       | 40,241 r/     | 42,000     |
| Talc                                           | 4,252      | 7,238        | 7,139         | 1,972 r/      | 2,000      |
| Turkey e/                                      | 4,000      | 4,000        | 4,000         | 5,000 r/      | 5,000      |
| United Kingdom: Talc, soapstone, pyrophyllite  | 4,298      | 5,322        | 5,500 e/      | 5,000 e/      | 5,000      |
| United States:                                 |            |              |               |               |            |
| Pyrophyllite                                   | W          | W            | W             | W             | W          |
| Talc                                           | 1,060,000  | 994,000      | 1,050,000     | 971,000 r/    | 925,000 7/ |
| Uruguay: Talc, soapstone, pyrophyllite         | 1,000 e/   | 898          | 1,133         | 972 r/        | 950        |
| Zambia: Talc e/                                | 80         | 80           | 80            | 80            | 80         |

See footnotes at end of table.

TABLE 7--Continued  
TALC AND PYROPHYLLITE: WORLD PRODUCTION, BY COUNTRY AND PRODUCT 1/ 2/

(Metric tons)

| Country 3/     | 1995         | 1996         | 1997          | 1998         | 1999 e/   |
|----------------|--------------|--------------|---------------|--------------|-----------|
| Zimbabwe: Talc | 2,080        | 1,076        | 1,023 r/      | 1,039 r/     | 1,000     |
| Grand total    | 8,490,000 r/ | 9,880,000 r/ | 10,400,000 r/ | 9,430,000 r/ | 9,470,000 |
| Of which:      |              |              |               |              |           |
| Pyrophyllite   | 2,000,000    | 1,950,000    | 2,390,000 r/  | 1,790,000 r/ | 1,790,000 |
| Steatite       | 714,000      | 712,000 r/   | 684,000 r/    | 714,000 r/   | 710,000   |
| Talc           | 2,420,000 r/ | 2,260,000 r/ | 2,320,000     | 2,220,000 r/ | 2,190,000 |
| Unspecified    | 3,360,000 r/ | 4,960,000 r/ | 5,040,000 r/  | 4,710,000 r/ | 4,780,000 |

e/ Estimated. r/ Revised. W Withheld to avoid disclosing company proprietary data; not included in "Total." -- Zero.

1/ World totals, U.S. data, and estimated data are rounded to no more than three significant digits; may not add to totals shown.

2/ Table includes data available through April 27, 2000.

3/ In addition to the countries listed, the former Czechoslovakia produces talc, but information is inadequate to make reliable estimates of output level.

4/ As reported in the Sumário Mineral 1999-2000.

5/ Data based on Iranian fiscal year beginning March 21 of year stated.

6/ Data based on Nepalese fiscal year beginning mid-July of year stated.

7/ Reported figure.

# Employment Preferences

## Chief Dentist, Correctional Facility

094199-00109344-6ceca

This multi-level recruitment is for:

094199-00109344-6ceca CHIEF DENTIST, CORRECTIONAL FACILITY

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Last Name

First Name

DOB Month

DOB Day:

Last four digits of SSN or other ID

First three letters of last name at birth

Email Address (if willing to accept email communication)

☐

Check here if this is a new email address

Mailing Address

City

State

Zip Code

☐

Check here if this is a new mailing address

**Only provide the following phone numbers if it is acceptable to call**

Home Phone

Work Phone

Alternate Phone

**Please complete the following employment preference information:**

You may pick one or more locations.

| Select | Location                        |
|--------|---------------------------------|
|        | Avenal State Prison             |
|        | CA State Prison,Sacramento      |
|        | Cal Correctional Institution    |
|        | Cal State Prison, San Quentin   |
|        | Cal Subst Abuse Trtmnt Fac      |
|        | Calif Institution For Men       |
|        | Calif Institution For Women     |
|        | Calif Rehabilitation Center     |
|        | Calif State Prison, Corcoran    |
|        | Calif State Prison, LA County   |
|        | Calif State Prison, Solano      |
|        | California Correctional Center  |
|        | California Medical Facility     |
|        | California Mens Colony          |
|        | Calipatria State Prison         |
|        | Centinela State Prison          |
|        | Central Calif Womens Facility   |
|        | Chaderjian School               |
|        | Chuckawalla Valley State Prison |
|        | Correction Central Region       |
|        | Correction Northern Region      |
|        | Correction Southern Region      |
|        | Correction Upper Northern Reg   |
|        | Correctional Trng Facility      |
|        | Deuel Vocational Institution    |
|        | Dewitt Nelson Training Center   |
|        | El Paso De Robles School        |
|        | Folsom State Prison             |

| Select | Location                      |
|--------|-------------------------------|
|        | Heman G. Stark YCF            |
|        | High Desert State Prison      |
|        | Ironwood State Prison         |
|        | Kern Valley St Prison         |
|        | Mule Creek State Prison       |
|        | North Kern State Prison       |
|        | Northern Youth Center, Stkn   |
|        | O H Close School, Stockton    |
|        | Pelican Bay State Prison      |
|        | Pine Grove Camp - Amador      |
|        | Pleasant Valley State Prison  |
|        | Preston Youth Cor Facility    |
|        | R J Donovan Corr Facility     |
|        | Richard A Mcgee Corr Trg Ctr  |
|        | Sacramento                    |
|        | Sacramento County             |
|        | Salinas Valley State Prison   |
|        | Sierra Conservation Center    |
|        | Southern Youth CRC            |
|        | Valley State Prison For Women |
|        | Ventura Youth Corr Facility   |
|        | Wasco State Prison            |

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Please select at least one item from each column to indicate conditions of employment your willing to accept:

- ☐ Permanent Full-time
- ☐ Permanent Part-time
- ☐ Permanent Intermittent
- ☐ Limited Term Full-time
- ☐ Limited Term Part-Time
- ☐ Limited Term Intermittent

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**Additional Options:**

- ☐ If you are currently eligible and wish to become inactive for this recruitment, please check here
- ☐ If you have previously inactivated yourself for this recruitment and would like to reactivate your application, please check here
- ☐ If you have never been eligible, and wish to withdraw from this recruitment, please check here

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Signature: \_\_\_\_\_ Date: \_\_\_\_\_